

```
import java.util.Scanner;
```

```
public class TriageSystem {
```

```
    static final int MAX_PATIENTS = 50;
```

```
    static String[] names = new String[MAX_PATIENTS];
```

```
    static int[] levels = new int[MAX_PATIENTS];
```

```
    static int n = 0;
```

```
    // Module 3: score from vitals
```

```
    static int calcScore(int hr, int spo2,
```

```
        double temp) {
```

```
        int score = 0;
```

```
        score += (hr > 110) ? 2 : 0;
```

```
        if (temp >= 38 && spo2 < 94) score++;
```

```
        if (spo2 < 90) score += 3;
```

```
        return score;
```

```
    }
```

```
    // Module 2: score -> triage level
```

```
    static int calcLevel(int score, int spo2,
```

```
        boolean conscious) {
```

```
        int level;
```

```
        if (score >= 9) level = 1;
```

```
        else if (score >= 6) level = 2;
```

```
        else if (score >= 4) level = 3;
```

```
        else if (score >= 2) level = 4;
```

```
        else level = 5;
```

```
        if (spo2 < 90 && !conscious) level = 1;
```

```
        return level;
```

```
}
```

```
// Module 3: overloaded print
```

```
static void print(String name, int lvl) {
```

```
    System.out.printf("%-10s L:%d%n", name, lvl);
```

```
}
```

```
static void print(String nm, int lvl, int wait) {
```

```
    System.out.printf("%-10s L:%d W:%d%n", nm, lvl, wait);
```

```
}
```

```
public static void main(String[] args) {
```

```
    Scanner sc = new Scanner(System.in);
```

```
    int choice = 1;
```

```
    while (choice != 0) {    // menu loop
```

```
        System.out.print("Name: ");
```

```
        String name = sc.nextLine();
```

```
        int hr;
```

```
        do {                // validate
```

```
            hr = sc.nextInt();
```

```
        } while (hr < 20 || hr > 250);
```

```
        int spo2 = sc.nextInt();
```

```
        double temp = sc.nextDouble();
```

```
        boolean con = sc.nextBoolean();
```

```
        int s = calcScore(hr, spo2, temp);
```

```
        int lvl = calcLevel(s, spo2, con);
```

```
        names[n] = name; levels[n] = lvl; n++;
```

```
        choice = sc.nextInt(); sc.nextLine();
```

```
}
```

```
for (int i = 0; i < n; i++) // for loop
```

```
    print(names[i], levels[i]);  
    int[] waitTimes = {12,45,8,30,22}; // 1D array  
    int sum = 0, longest = 0;  
    for (int w : waitTimes) {  
        sum += w;  
        if (w > longest) longest = w;  
    }  
    double avg = (double) sum / waitTimes.length;  
    System.out.printf("Avg %.1f Max %d%n", avg, longest);  
}  
}
```